

Requirements and load cases for the seam

Fall milestones 1 and 2, due 27 September. Manos Theofanous for Prof. Rick Fleeter, ENGN 1972. 25 September 2026.

Fifteen requirements, each with where it comes from. "Seam" is where two panels meet: two aluminium edge frames bolted face to face, two gaskets between them. Numbers marked as placeholders get checked against the named standard before the week 8 standards memo.

#	Requirement	Source	How it's verified
R1	Hold 1 atm (101.3 kPa) cabin against vacuum without failure at 2.0 times limit load	TRUTH App. A, 2.4.2 (prototype ultimate factor)	FEA; T1 to 2 atm
R2	No yield in the frames, ribs or bolts at proof, 1.5 times limit (placeholder)	NASA-STD-5001B, to confirm	FEA; T1, T2
R3	Carry the seam tension of 276.7 kN/m (LC1) with the inner-gasket gap well under the 1.33 mm design squeeze	Derived, Sheet 5; gland per Rev C (25 % squeeze of 5.33 cord)	T1 clip gauges
R4	Dual seal: each gasket holds 1 atm alone; the volume between them can be pressurised and monitored; 30 min hold with no measurable decay	TRUTH 2.4.5, 10.2	T1 leak check; 550 mm article
R5	Inner pane (L7) deflection at most 1/200 of the pane radius at 1 atm (inscribed radius, 613 mm, the conservative reading)	TRUTH 9.5	Hand calc: 7.2 mm against 3.1 mm, fails if simply supported. FEA, T2
R6	Inner pane stress at most the sustained-load allowable at a stated probability of failure, from measured Weibull data scaled to 2.19 m2	TRUTH 9.5; NASA-STD-5018; ASTM C1239	T0, including a slow-crack-growth set (ASTM C1368), plus analysis (item 4)
R7	Frame deformation at the panel perimeter at most 1 mm under pressure and thermal load	TRUTH 9.5	FEA; T2
R8	Keep seal squeeze and bolt preload over -40 to +120 C, 90 min orbit	TRUTH 2.4.2, 10.3	FEA later; spring thermal test if a chamber is found
R9	Survive 10,000 cycles of 0 to 1 atm without a change in leak rate	TRUTH 8.7, 10.6	Spring 2027 onward
R10	Shear key accepts 5 mm in-plane misalignment during pull-in	TRUTH 10.7	Rev D geometry; dry fit
R11	Fasteners set from the panel exterior with one-side access	TRUTH 10.7	Rev D; parts memo (item 7)
R12	Every fastener and seal designation resolves to a real part at its issuing body	Team rule, 15 Sep; NASM/MS listings	Parts memo (item 7)
R13	First panel mode above 50 Hz	TRUTH 9.5	Modal FEA
R14	Material properties from MMPDS: 6061-T6 for the prototype, 2219-T851 for flight	TRUTH 10.2; item 3	Materials pick (4 Oct)
R15	Every test has model predictions filed before loading, and passes within 10 percent in the linear range	Test plan Rev B	Each test report

Load cases, reconciled

The March plan had 99 kN/m and my earlier note had 282 kN/m. Both were real. 99 kN/m is the push of the panel on its edge (V). Because two flat panels meet at 138.19 degrees, each also pulls in its own plane ($N = 258$ kN/m), and V and N add to $T = 276.7$ kN/m straight across the joint, which is $p R / 2$ for the sphere through the edge midpoints. T is what the bolts carry if the panels carry N. It is an upper bound: if the in-plane load runs along the seams to the corners instead, the pull across the joint drops toward $V \sin 20.9 \text{ deg} = 35$ kN/m. The load cases below use the bound until the FEA settles the split.

Case	Seam load	Per 75 mm bolt	550 mm article	Source
LC1 limit, 1 atm	276.7 kN/m	20.7 kN	152 kN	TRUTH App. A; derived
LC2 design, x 1.2 transient (docking about 20 %)	332.0 kN/m	24.9 kN	183 kN	TRUTH 2.4.2
LC3 proof, 1.5 x LC2	498.0 kN/m	37.3 kN	274 kN	Placeholder, NASA-STD-5001B
LC4 ultimate, 2.0 x LC2	664.0 kN/m	49.8 kN	365 kN	TRUTH 2.4.2
LC5 leak check	1 atm between the gaskets	-	-	TRUTH 2.4.5
LC6 thermal	-40 to +120 C, 90 min orbit	-	-	TRUTH 2.4.2
Old basis, V only (superseded)	98.7 kN/m	7.4 kN	54.3 kN	TRUTH 10.3

What changes because of this

- At the bound, the -10 bolt (about 34 kN on the shank area; the NASM rated value replaces it in item 7) has 1.6 at LC1, 0.9 at proof and 0.7 at ultimate. Bolt size and pitch are the first Rev D decision.
- The 550 mm test article is a 150 to 365 kN test at the bound, so it can't be pulled in a 100 kN frame. It gets the dry fit and the leak test; the pull test moves to short slices (T1).
- The TRUTH 2.4.2 coupon loads (54.3, 65.2, 130.3 kN) and 10.3 (7.4 kN per bolt, 5.2 times margin) need correcting in TRUTH. I'll bring the wording to you before I change it.
- Open at shell scale: whether the 258 kN/m in-plane load stays in the panels or runs along the seams to the corners (test plan, Sheet 6). The FEA size study settles it.